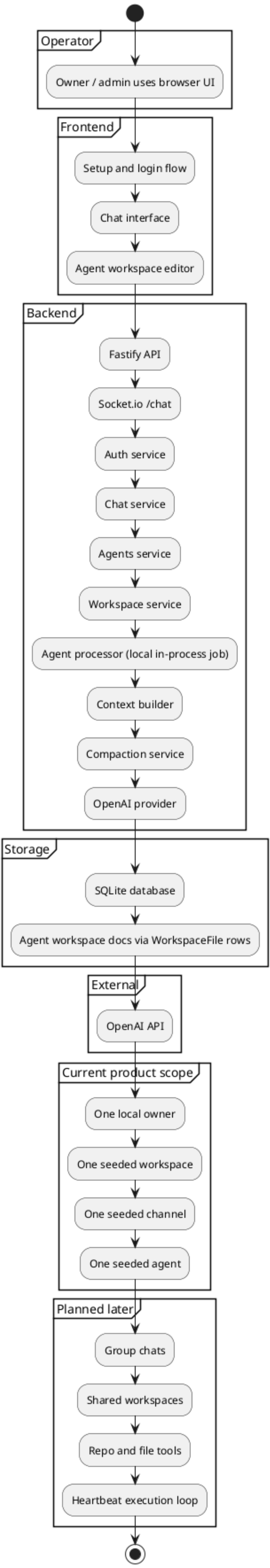
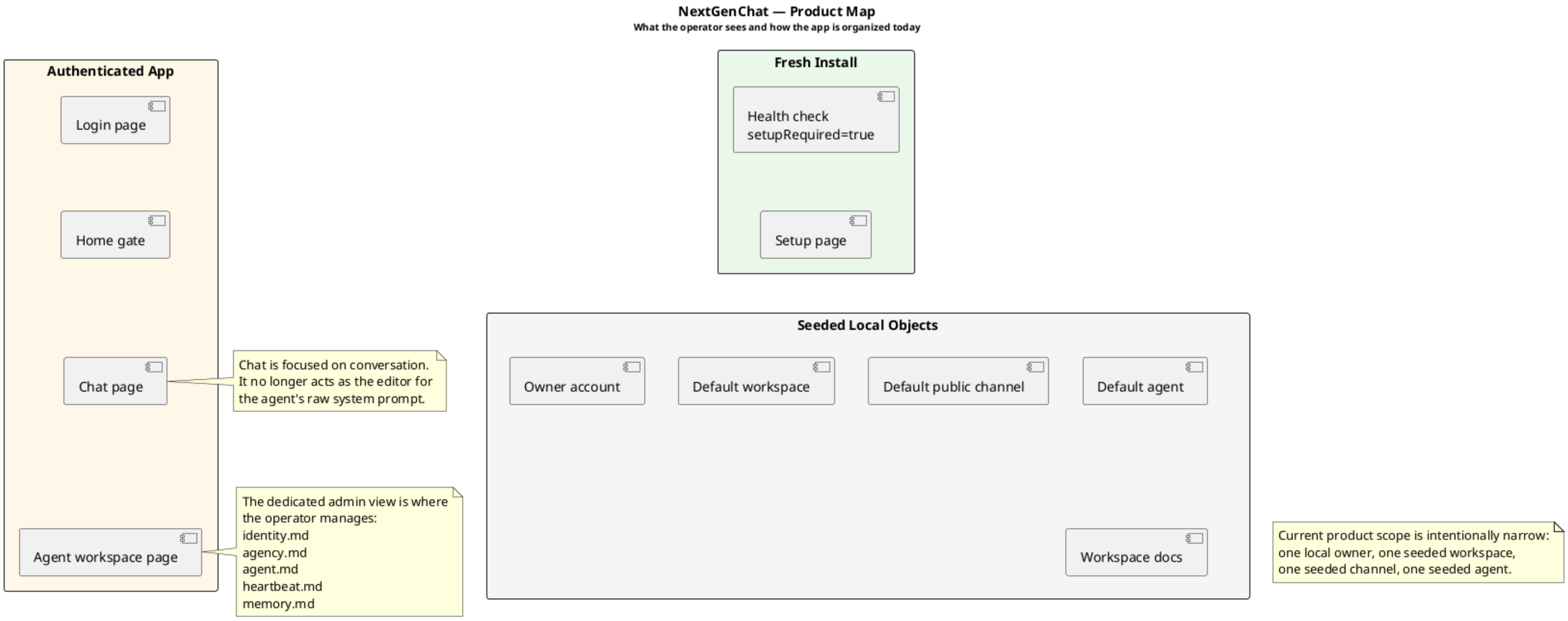
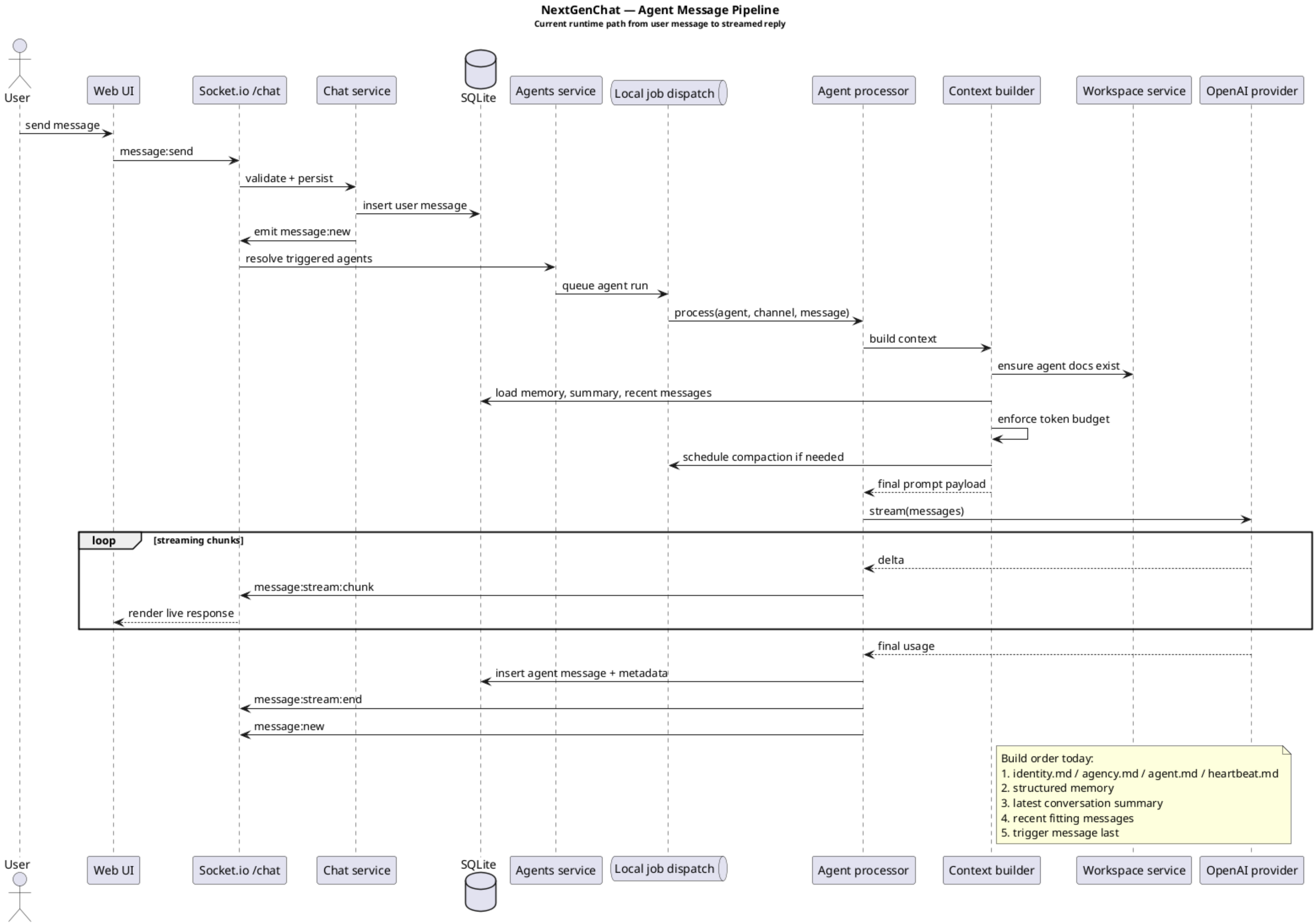
NextGenChat — System Overview

Current local-first runtime and major subsystems

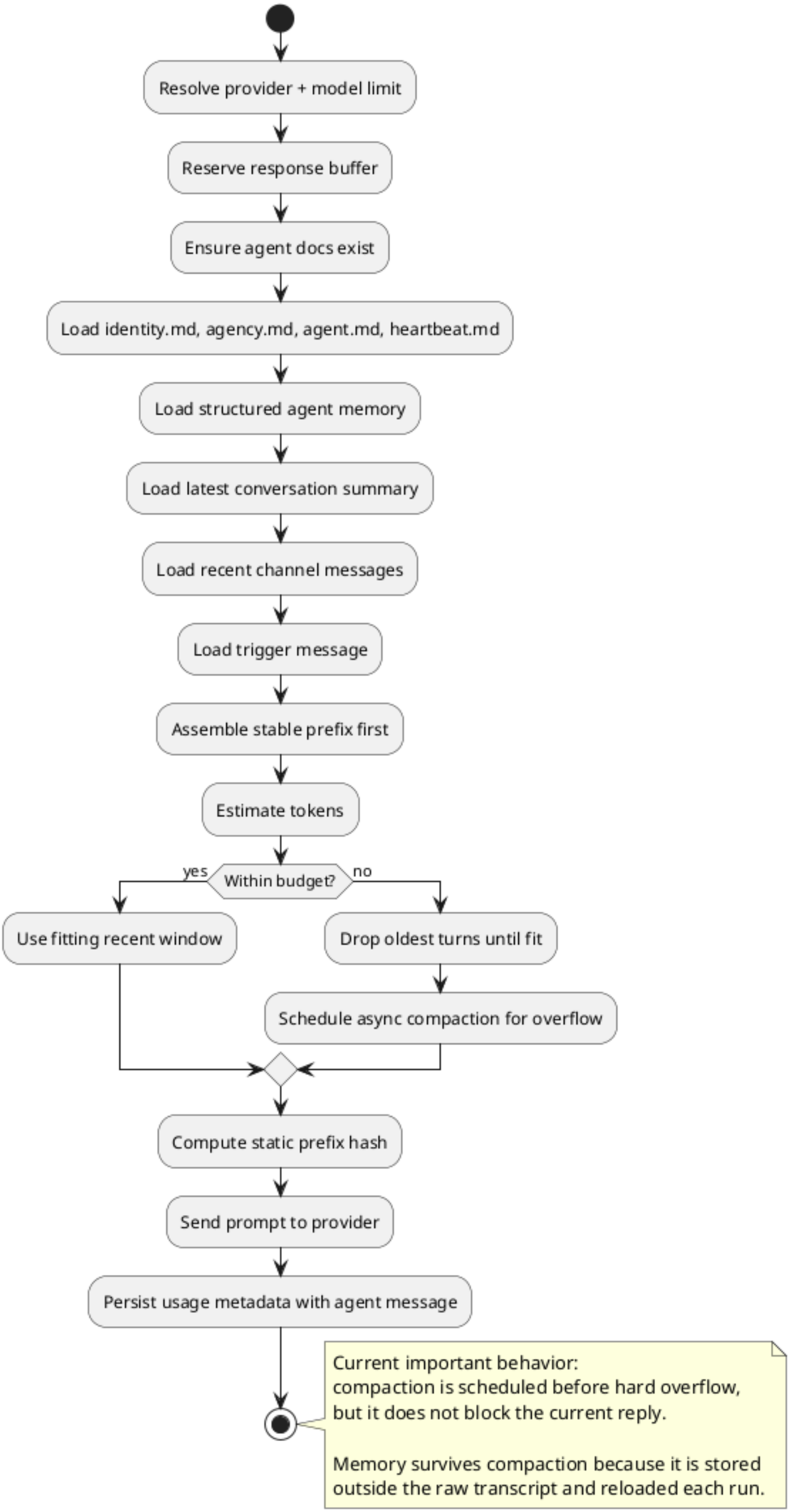


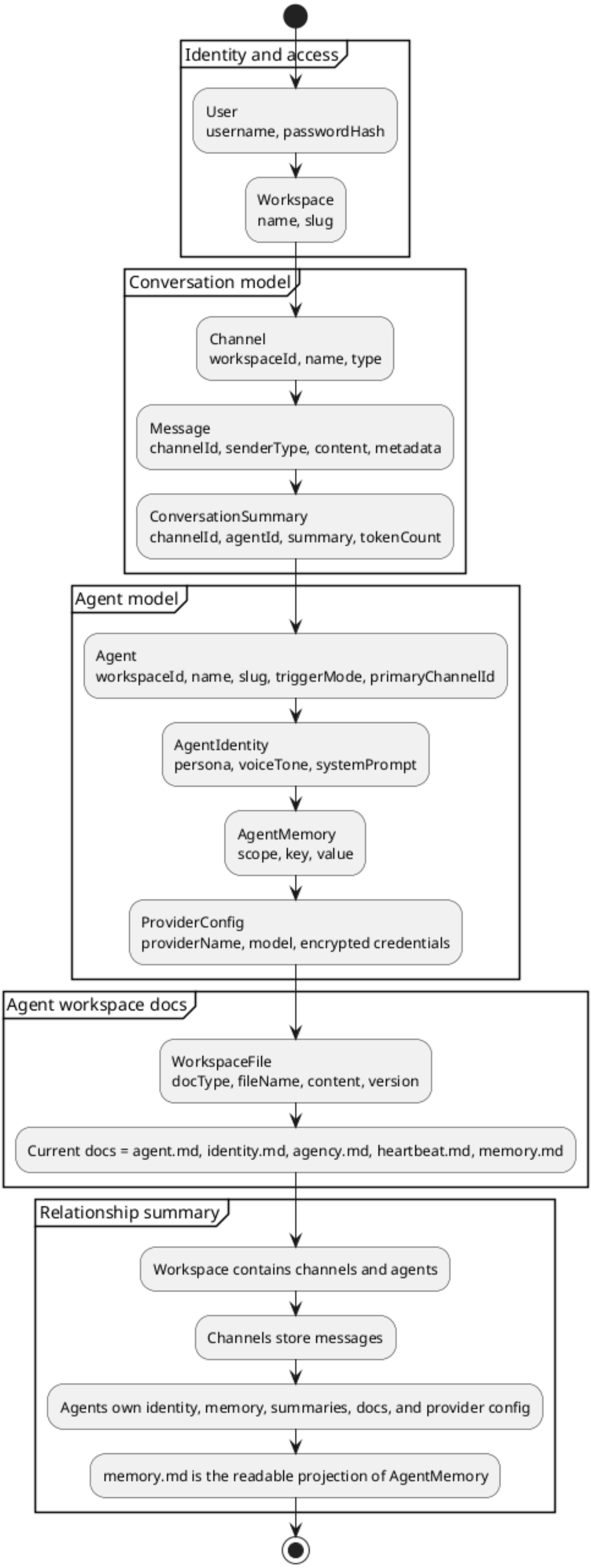

NextGenChat — Product Map
What the operator sees and how the app is organized today
Authenticated App
Login page
Home gate
Chat page
Agent workspace page
Chat is focused on conversation. It no longer acts as the editor for the agent's raw system prompt.
The dedicated admin view is where the operator manages:
identity.md
agency.md
agent.md
heartbeat.md
memory.md
Fresh Install
Health check
setupRequired=true
Setup page
Seeded Local Objects
Owner account
Default workspace
Default public channel
Default agent
Workspace docs
Current product scope is intentionally narrow:
one local owner, one seeded workspace,
one seeded channel, one seeded agent.


NextGenChat — Agent Message Pipeline
Current runtime path from user message to streamed reply
User
Web UI
Socket.io /chat
Chat service
SQLite
Agents service
Local job dispatch
Agent processor
Context builder
Workspace service
OpenAI provider
send message
message:send
validate + persist
insert user message
emit message:new
resolve triggered agents
queue agent run
process(agent, channel, message)
build context
ensure agent docs exist
load memory, summary, recent messages
enforce token budget
schedule compaction if needed
final prompt payload
stream(messages)
loop
[streaming chunks]
delta
message:stream:chunk
render live response
final usage
insert agent message + metadata
message:stream:end
message:new
Build order today:
1. identity.md / agency.md / agent.md / heartbeat.md
2. structured memory
3. latest conversation summary
4. recent fitting messages
5. trigger message last

NextGenChat — Context Model

How prompt assembly, compaction, and cache-sensitive structure work today



NextGenChat — Current Domain Model

Core persisted entities in the running app